

YOLO11n-seg ONNX vs .pt Validation and Offline Comparison

Computer Vision Subteam - Sprint 1

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1. Purpose

This report compares the custom YOLO11n-seg ONNX inference pipeline with the existing Ultralytics .pt pipeline. The purpose is to check whether the ONNX implementation is producing similar post-processing results before the pipeline is taken further into ROS2 integration. The .pt pipeline is used here as the implementation reference; this is not a comparison against ground-truth annotations.

The comparison is directly relevant to Sprint 1 task CV-06, which requires the model output to be decoded in original image coordinates and compared with the existing evaluation code. It also provides part of the offline evidence for CV-09. This report does not claim that all CV-09 checks are complete.

2. Comparison setup

The final ONNX CSV and .pt CSV each contained 26 matching image names, with no unmatched rows. The same image name was used to pair the two outputs.

- Confidence was compared using the absolute difference between the ONNX and .pt confidence values.
- MAR centre was compared using Euclidean pixel distance between ONNX mar_center_x/mar_center_y and the .pt minimum-area-rectangle centre stored as centroid_x/centroid_y.
- Orientation was compared modulo 180 degrees because the long-axis direction is equivalent after a 180-degree reversal.
- MAR side lengths were sorted into short and long sides before comparison because OpenCV may swap the width and height fields depending on the rectangle orientation.

The ONNX pipeline reconstructs the segmentation mask manually from the ONNX outputs, while the reference script obtains the mask through the Ultralytics .pt pipeline. Both then use a minimum-area rectangle to obtain centre, side lengths and long-axis orientation.

3. Quantitative results

Measurement	Mean difference	Median difference	Largest difference
Confidence	0.0065	0.0026	0.0445
MAR centre	12.86 px	8.03 px	54.04 px
Long-axis angle	0.22°	0.03°	0.98°
Short MAR side	5.03%	5.52%	10.83%
Long MAR side	1.50%	0.15%	8.79%

Main observations. 21/26 images had a confidence difference below 0.01. 19/26 MAR centres were within 15 pixels and 22/26 were within 20 pixels. All 26/26 angle differences were below 1 degree, with 21/26 at or below 0.5 degree.

MAR dimensions. The long side was generally very close between the two pipelines, with a mean absolute difference of 1.50%. The short side showed a larger difference, averaging 5.03%. Across the full set, the signed short-side difference was +4.94% on average, meaning the ONNX MAR short side tended to be slightly larger than the .pt result. The corresponding signed long-side difference was +0.64%.

3.1 Selected matched examples

Image	Conf. diff	Centre diff	Angle diff	Short-side diff	Long-side diff
5.jpg	0.0373	6.71 px	0.00°	1.85%	0.25%
5_aug_9.jpeg	0.0113	4.12 px	0.05°	2.11%	0.10%
7_aug_4.jpeg	0.0009	19.03 px	0.63°	10.83%	0.20%
88_aug_2.jpeg	0.0008	5.10 px	0.20°	9.54%	0.06%

4. Visual comparison

The four matched examples below use the same source image for both pipelines. The ONNX output is shown on the left and the .pt reference output on the right. The comparison focuses on mask placement, the green minimum-area rectangle and the orange long-axis orientation.

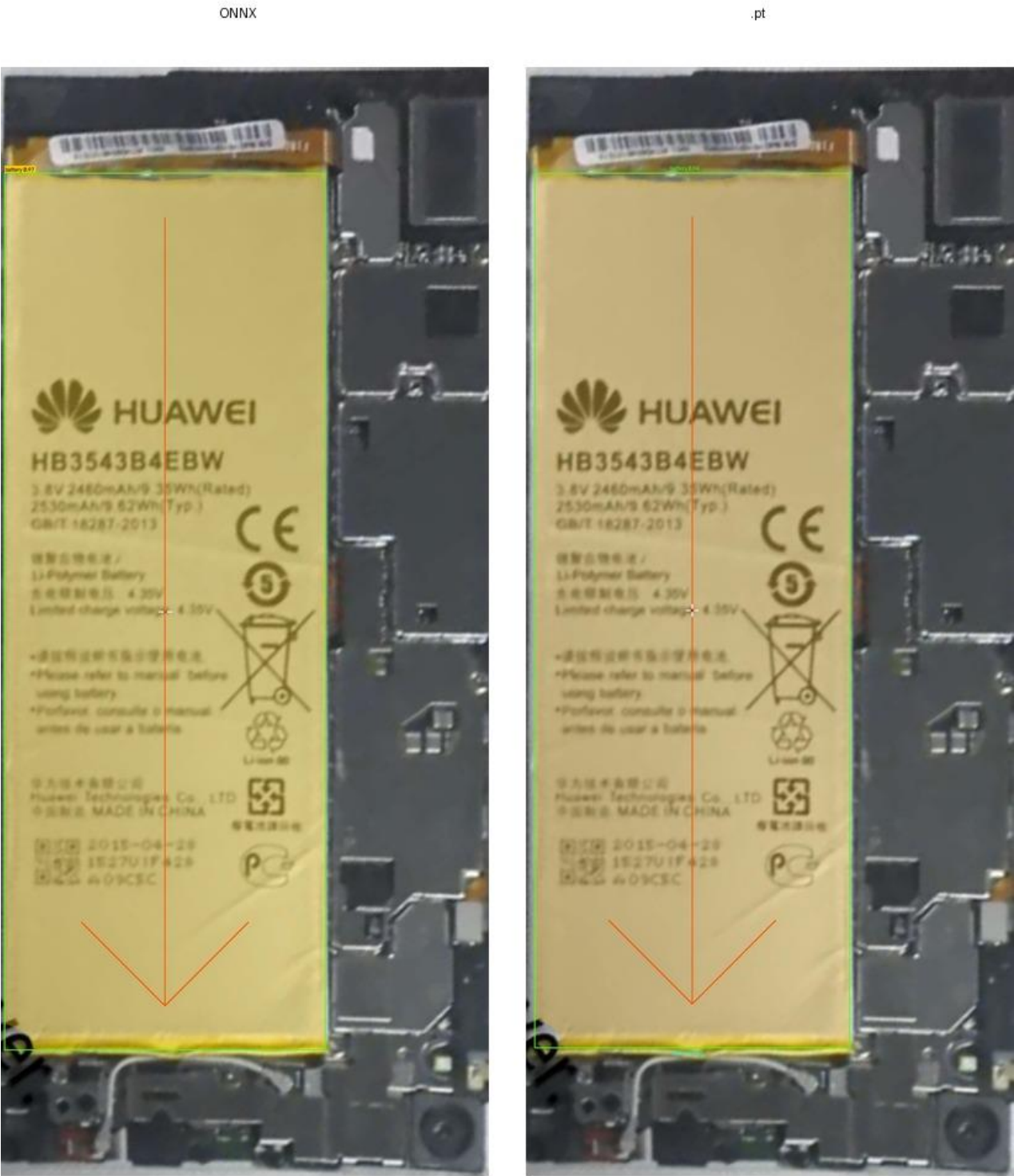


Figure 1. ONNX (left) and .pt (right): CamScanner_08-09-2025_06.15_5.jpg

This pair has the same 90° long-axis angle in both outputs. The MAR centre differs by 6.71 pixels and the long-side difference is 0.25%.

ONNX

.pt



Figure 2. ONNX (left) and .pt (right): CamScanner_08-09-2025_06.15_5_aug_9.jpeg

The rotated example remains closely aligned: the angle difference is 0.05°, the centre difference is 4.12 pixels, and the long-side difference is 0.10%.

ONNX

.pt

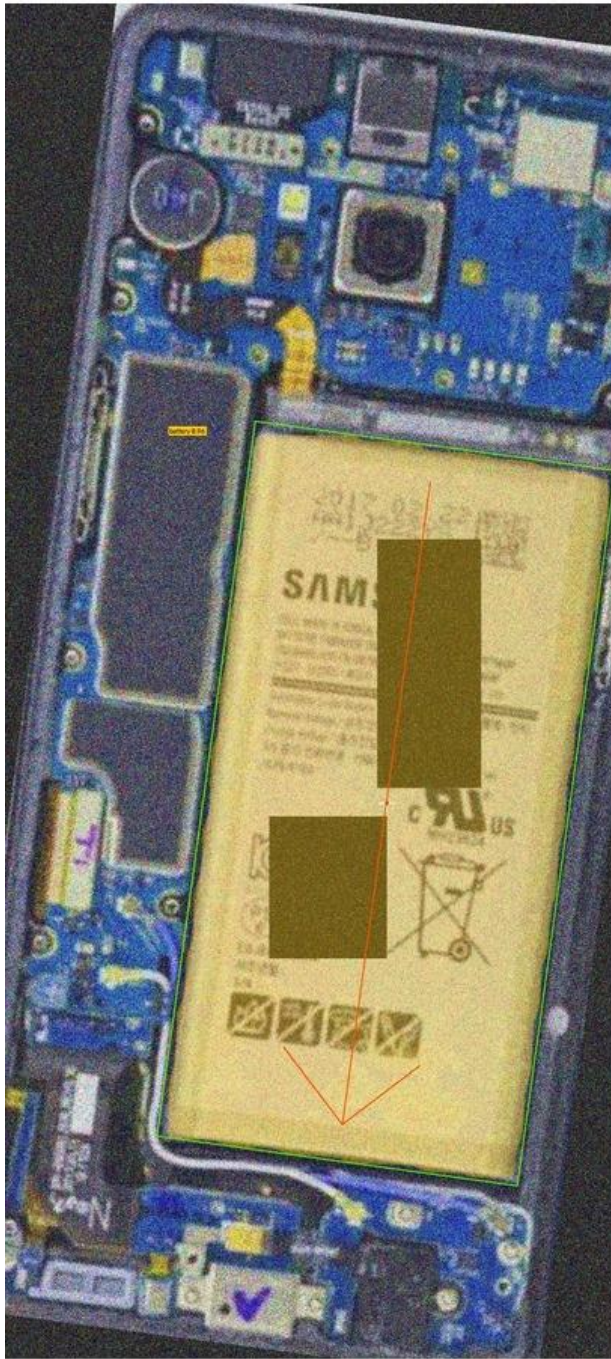


Figure 3. ONNX (left) and .pt (right): CamScanner_08-09-2025_06.15_7_aug_4.jpeg

This augmented example shows one of the larger short-side differences (10.83%), while the angle difference is 0.63° and the long-side difference is only 0.20%.

ONNX

.pt

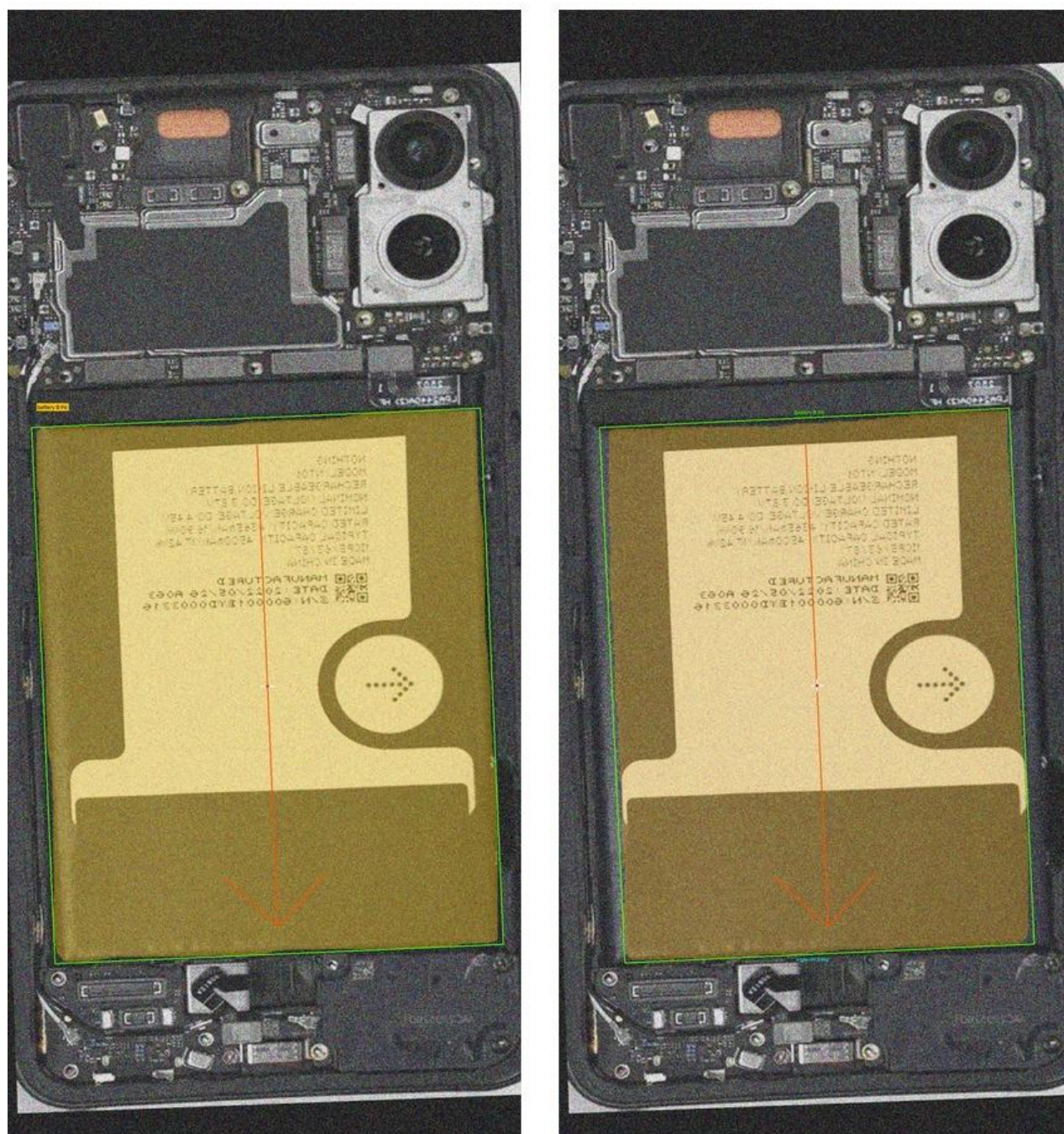


Figure 4. ONNX (left) and .pt (right): CamScanner_08-09-2025_06.15_88_aug_2.jpeg

The two outputs give a very similar long-axis result: the angle difference is 0.20° and the long-side difference is 0.06%. The short-side difference is 9.54%.

5. Discussion

The strongest agreement is in the long-axis orientation. The mean angle difference is only 0.22° , and the largest difference across the 26 matched images is 0.98° . This shows that the minimum-area-rectangle orientation logic in the ONNX pipeline is behaving very similarly to the .pt reference on this test set.

Confidence values are also close overall. The largest confidence difference occurs on CamScanner_08-09-2025_06.15_165_aug_1.jpeg, where the absolute difference is 0.0445. This is larger than most of the set but does not change the fact that both pipelines detect the battery in the supplied result.

The remaining consistent difference is the short side of the minimum-area rectangle. The ONNX output is slightly larger on this dimension on average. The largest short-side percentage difference in the test set is 10.83% on CamScanner_08-09-2025_06.15_7_aug_4.jpeg. In contrast, the long side is usually much closer. These measurements are recorded as an observed difference rather than being adjusted artificially to force the two pipelines to match.

The visual examples support the numerical comparison: both pipelines identify the same battery region and produce similar rotated boxes and long-axis directions. The comparison therefore provides evidence that the custom ONNX post-processing is consistent with the existing .pt implementation for the outputs tested here.

6. Sprint 1 relevance and current scope

This work directly supports CV-06 (Detection Post-processing), particularly the requirement to recover battery geometry in original image coordinates and compare the result with the existing evaluation code. It also contributes offline evidence toward CV-09 (Offline Pipeline Testing).

The current evidence does not include ROS2 node stability, empty/no-battery frame behaviour, video or rosbag testing, or measured FPS/latency. Those items are therefore not reported as completed here. They should be tested separately when the ROS2 pipeline is available.

7. Conclusion

Across the 26 matched images, the ONNX and .pt pipelines produced closely aligned results for confidence, MAR centre, long-axis orientation and long-side geometry. The average angle difference was 0.22° and the average MAR-centre difference was 12.86 pixels. The main remaining difference was the short MAR side, which differed by 5.03% on average and was generally larger in the ONNX output.

Based on the supplied offline test set, the ONNX post-processing behaves consistently enough with the .pt reference to be used as the current implementation for the next integration stage, while keeping the short-side difference documented for later testing.